

Renal and Systemic Vasculitis

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DEFINITION

The large number and variety of kidney vessels make the kidneys targets for a variety of systemic vasculitides, especially those affecting small vessels.¹⁻⁴ Vasculitis involving the kidneys produces a variety of clinical manifestations, depending on the type of kidney vessel affected. Vasculitides are categorized as large-, medium-, and small-vessel vasculitis (Figs. 26.1 and 26.2). The 2012 Chapel Hill Consensus Conference definitions are used throughout this chapter (Table 26.1).⁴

Several vasculitides in Fig. 26.2 are reviewed in other chapters and are discussed here only in the context of differential diagnosis. Nephrologists often encounter patients with small-vessel vasculitides because these frequently cause glomerulonephritis (GN). Therefore, small-vessel vasculitis (SVV) is the primary focus of this chapter.

Small-Vessel Vasculitis

SVV is necrotizing vasculitis affecting predominantly vessels that are smaller than arteries, including capillaries, venules, and arterioles; however, small arteries can be involved.⁴ The most common renal target for SVV is glomerular capillaries and the most common kidney clinical manifestation is GN.

Medium-Vessel Vasculitis

Medium-vessel vasculitis is necrotizing arteritis affecting predominantly major visceral arteries.⁴ In the kidneys, interlobar arteries and arcuate arteries are most affected, although any arteries from the main renal artery to the smallest interlobular arteries may be affected. Arterial inflammation and necrosis may result in thrombosis or rupture, causing kidney infarction and hemorrhage, respectively.

Large-Vessel Vasculitis

Large-vessel vasculitis is chronic granulomatous arteritis affecting the aorta and its major branches more often than other forms of vasculitis.⁴ With kidney involvement, the ostia of the renal arteries and the main renal arteries are most often affected. The most common clinical kidney manifestation is renovascular hypertension.

SMALL-VESEL PAUCI-IMMUNE VASCULITIS

SVV can be divided into *immune complex* SVV with moderate to marked vessel wall deposits of immunoglobulin and *pauci-immune* SVV with few or no immune deposits in vessel walls.⁴ Pauci-immune SVV often is associated with circulating antineutrophil cytoplasmic autoantibodies (ANCA).¹ The ANCA-associated vasculitides share an indistinguishable form of necrotizing SVV that affects capillaries, venules, arterioles, and small arteries.¹⁻⁵ Some patients with ANCA-associated vasculitis (AAV) have no evidence of arterial involvement,

although they have involvement of glomerular capillaries, causing GN; pulmonary alveolar capillaries, causing pulmonary hemorrhage; or dermal venules, causing purpura. The clinicopathologic variants of pauci-immune SVV are categorized based on clinical, laboratory, and pathologic findings, as follows⁴:

- *Microscopic polyangiitis* (MPA) is pauci-immune SVV in the absence of evidence for necrotizing granulomatous inflammation.
- *Granulomatosis with polyangiitis* (GPA) is pauci-immune vasculitis associated with necrotizing granulomatous inflammation, most often affecting the respiratory tract.
- *Eosinophilic granulomatosis with polyangiitis* (EGPA; formerly called Churg-Strauss syndrome) is pauci-immune vasculitis associated with asthma, eosinophilia, and necrotizing granulomatous inflammation.

MPA, GPA, and, less frequently, EGPA share an indistinguishable pattern of GN that is expressed as vasculitis in glomerular capillaries.¹ In the acute phase, the GN usually has necrosis and crescents, an absence or paucity of immunoglobulin deposition, and is designated *pauci-immune crescentic glomerulonephritis*. In the absence of systemic vasculitis, pauci-immune crescentic GN is sometimes referred to as *renal-limited vasculitis*.

Pathogenesis

MPA, GPA, EGPA, and renal-limited pauci-immune crescentic GN are all associated with ANCA.⁴⁻⁶ The most common antigen targets of ANCA are proteinase 3 (PR3) and myeloperoxidase (MPO).^{7,8}

The association of ANCA with this form of SVV suggests that ANCA are involved in disease pathogenesis.⁴⁻⁶ Correlation of ANCA titers with disease activity in some individuals suggests a pathogenic role; however, this correlation is not strong and some patients with clinically and pathologically typical MPA, GPA, or renal-limited pauci-immune crescentic GN are negative using conventional serologic ANCA testing. MPO-ANCA epitope specificity determines the pathogenicity, detectability, and clinical predictive value of circulating MPO-ANCA.⁹ For example, ANCA with certain epitope specificities occur only during active disease, whereas other MPO-ANCA epitope specificities occur regardless of disease status and in healthy controls (natural ANCA), although at very low titers.^{10,11} Some patients with AAV who are negative by conventional serologic testing have MPO-ANCA with restricted epitope specificity that can be detected with unmasking techniques.¹⁰

The pathogenic potential of ANCA is supported by the observation that certain drugs, such as propylthiouracil, hydralazine, and penicillamine, can induce AAV.¹² Cocaine adulterated with levamisole can induce AAV with high titer MPO-ANCA, PR3-ANCA, and ANCA specific for elastase.¹² Levamisole-induced vasculitis has frequent cutaneous leukocytoclastic angiitis and upper respiratory tract involvement but rarely kidney or lung involvement.

Renal Vascular Involvement in Vasculitides

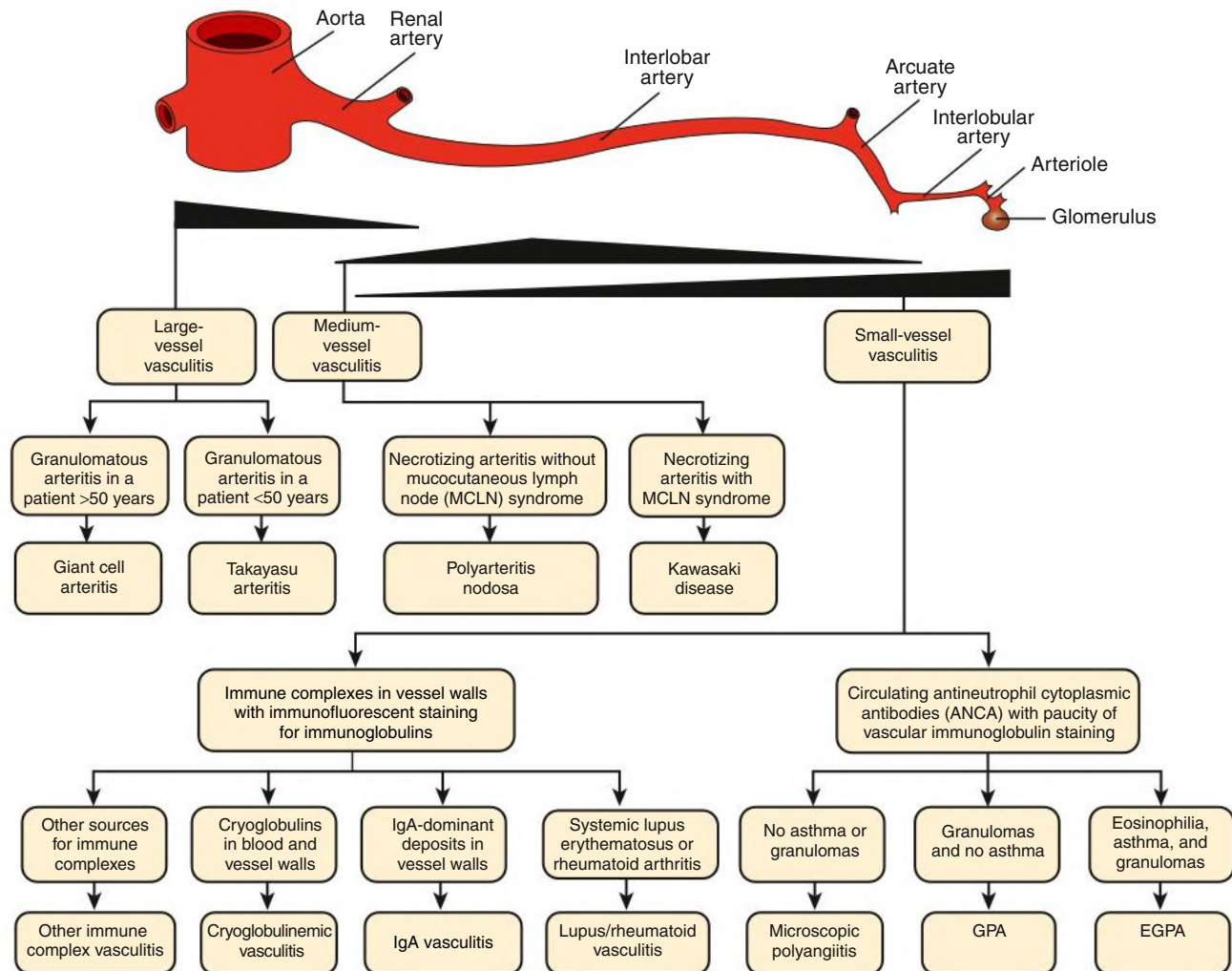


Fig. 26.1 Renal Vasculitis. Predominant distribution of kidney vascular involvement by a variety of vasculitides. The heights of the trapezoids represent the relative frequency of involvement of different portions of the renal vasculature by the three major categories of vasculitis. *EGPA*, Eosinophilic granulomatous polyangiitis; *IgA*, immunoglobulin A.

In vitro observations suggest mechanisms by which ANCAs cause vascular injury.^{5,6} Cytokine priming causes neutrophils to increase expression of ANCA antigens on their surfaces where ANCAs can bind. Cytokine-primed neutrophils activated by ANCAs release lytic enzymes from granules, generate toxic oxygen metabolites, and kill cultured endothelial cells. ANCA-antigen complexes adsorb onto endothelial cells where they could participate in *in situ* immune complex formation. ANCA activation of neutrophils is mediated by F(ab')₂ binding to neutrophils and to a greater degree by FcR engagement. These events cause vasculitis as a result of neutrophils adhering to, penetrating, and destroying vessel walls (Fig. 26.3).

The ability of ANCA to cause pauci-immune necrotizing and crescentic GN and vasculitis is demonstrated in animal models induced with MPO-ANCA. No widely accepted model of PR3-ANCA disease has been developed because of differences in murine PR3 expression and limited homology between mice and humans.⁶ Wild-type or immunodeficient mice intravenously (IV) injected with anti-MPO antibodies develop pauci-immune focal necrotizing crescentic GN.¹³

A rat model of pauci-immune necrotizing crescentic GN developed by immunizing rats with human MPO developed antibodies that cross-react with rat MPO.¹⁴ MPO-ANCA GN in mice is mediated by neutrophil activation, modulated by the FcγR repertoire, and is prevented by neutrophil depletion.^{6,13} Alternative complement pathway activation plays a role in amplifying ANCA-induced inflammation.¹⁵ ANCA-activated neutrophils release factors that activate the alternative complement pathway, resulting in C5a generation, which is chemotactic for neutrophils and primes for further activation by ANCAs.¹⁶ Patients with active AAV (as defined by the Birmingham Vasculitis Activity Score [BVAS]) have increased plasma levels of C3a, C5a, soluble C5b-9, and Bb, and plasma levels of Bb correlated with percentage of cellular crescents in kidney biopsy samples and BVAS.¹⁷ Further studies concluded that complement cascade activation differs with disease activity and serotype.¹⁸

Clinical and experimental data indicate that ANCAs can activate neutrophils and cause vasculitis, especially if there are synergistic proinflammatory stimuli. The synergistic neutrophil-priming

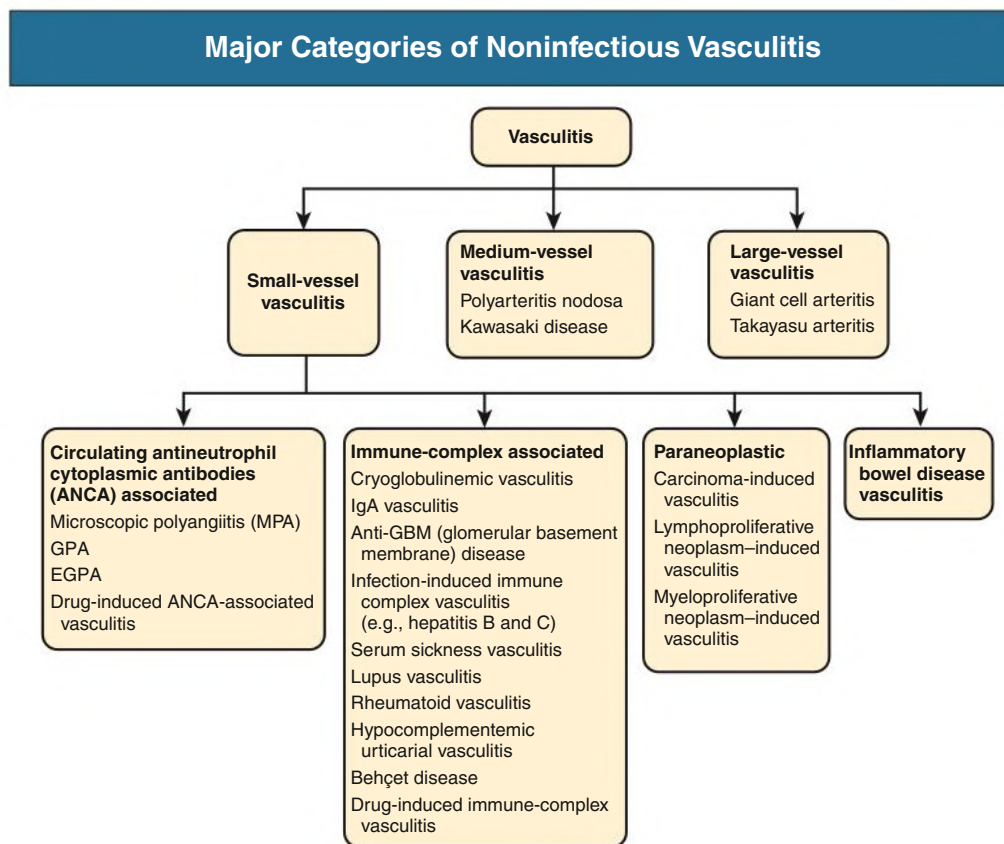


Fig. 26.2 Major Categories of Noninfectious Vasculitis. Not included are vasculitides that are known to be caused by direct invasion of vessel walls by infectious pathogens, such as rickettsial vasculitis and neisserial vasculitis. EGPA, Eosinophilic granulomatous polyangiitis; GPA, granulomatous polyangiitis.

inflammatory process may be reflected in the frequent association of AAV onset with an influenza-like syndrome.¹⁹

Initiation of the ANCA autoimmune response is less understood but involves human leukocyte antigen (HLA) with specific molecular recognition capabilities.^{20,21} Genome-wide association studies have identified associations between HLA alleles and AAV (HLA-DP for PR3-ANCA and HLA-DQ for MPO-ANCA).^{20–22} Because certain HLA have a greater predilection for binding auto-antigenic peptides, it is logical that HLA associated with AAV has a higher propensity to bind MPO or PR3 peptides. Within the MPO-ANCA serotype, a small number of MPO peptides bind specific HLA, and T cells react to these same peptides.¹¹ These autoreactive T cells are capable of shaping the B-cell response, leading to ANCA production.

Epidemiology

GPA, MPA, and EGPA occur at any age but usually begin during the fifth through seventh decades of life, with a peak incidence of 65 to 75 years. There is a slight male predominance. AAV has geographic and race/ethnicity differences in prevalence.²³ In the United Kingdom, GPA (148 per 1 million) is more common than MPA (65 per million), and MPA is more common than EGPA (46 per 1 million). In France, AAV prevalence is twice as high in Europeans (105 per 1 million) compared with non-Europeans (53 per million). In the United States, the incidence is disproportionately greater in Whites than in Blacks, which may be caused by HLA differences.²⁴

In the United Kingdom and northern Europe, PR3-ANCA (usually associated with GPA) is more common than MPO-ANCA; however, in Southern Europe, Asia, and India, MPO-ANCA and MPA are more common.²³ In the United States, there is a similar trend with more PR3-ANCA and GPA in northern states and more MPO-ANCA and MPA in southern states. Geographic and racial differences may be related to HLA differences.^{20,21}

Clinical Manifestations

Generalized nonspecific manifestations of systemic inflammatory disease (fever, malaise, anorexia, weight loss, myalgias, and arthralgias) often are present. Many patients trace the onset of their vasculitic disease to a flu-like illness.¹⁹

Clinical manifestations of GPA, MPA, and EGPA are extremely varied because they are influenced by the organs affected. The three categories of vasculitis share features caused by SVV, and patients with GPA and EGPA have additional defining features.^{4,25,26}

Kidney involvement occurs often in GPA and MPA and is uncommon in EGPA (Table 26.2). Most common kidney manifestations are hematuria, proteinuria, and kidney failure. Kidney failure often has the characteristics of rapidly progressive glomerulonephritis (RPGN) in GPA and MPA but is less severe in EGPA. A cohort of more than 300 patients with pauci-immune crescentic GN evaluated at kidney biopsy had a mean age of 56 years (range, 2–92 years), male-to-female ratio of 1.0:0.9, mean serum creatinine concentration of 6.5 mg/dL (range, 0.8–22.1 mg/dL; 69–1900 μ mol/L), and proteinuria of 1.9 g/day (range, 0.1–18g/day).²⁷

TABLE 26.1 Names and Definitions of Vasculitis

Category/Name	Definition
Large-Vessel Vasculitis	
Takayasu arteritis	Arteritis, often granulomatous, predominantly affecting the aorta and/or its major branches. Onset usually in patients younger than 50 years.
Giant cell arteritis	Arteritis, often granulomatous, usually affecting the aorta and/or its major branches, with predilection for branches of carotid and vertebral arteries; often involves temporal artery. Onset usually in patients older than 50 years and often associated with polymyalgia rheumatica.
Medium-Vessel Vasculitis	
Polyarteritis nodosa	Necrotizing arteritis of medium or small arteries without GN or vasculitis in arterioles, capillaries, or venules and not associated with ANCA.
Kawasaki disease	Arteritis associated with mucocutaneous lymph node syndrome and predominantly affecting medium and small arteries; coronary arteries are often involved; aorta and large arteries may be involved. Usually occurs in infants and young children.
Small-Vessel Vasculitis	
ANCA-Associated Small-Vessel Vasculitis	
Necrotizing vasculitis, with few or no immune deposits, predominantly affecting small vessels (capillaries, venules, arterioles, small arteries), associated with MPO-ANCA or PR3-ANCA. Not all patients have ANCA. Add prefix indicating ANCA reactivity (e.g., PR3-ANCA, MPO-ANCA, ANCA-negative).	
Microscopic polyangiitis (MPA)	Necrotizing vasculitis, with few or no immune deposits, predominantly affecting small vessels (capillaries, venules, arterioles). Necrotizing arteritis involving small and medium arteries may be present. Necrotizing GN is common. Pulmonary capillaritis often occurs. Granulomatous inflammation is absent.
Granulomatosis with polyangiitis (GPA)	Necrotizing granulomatous inflammation usually involving upper and lower respiratory tract and necrotizing vasculitis affecting predominantly small to medium vessels (capillaries, venules, arterioles, arteries, veins). Necrotizing GN is common.
Eosinophilic granulomatosis with polyangiitis (EGPA)	Eosinophil-rich and necrotizing granulomatous inflammation often involving the respiratory tract, and necrotizing vasculitis predominantly affecting small to medium vessels and associated with asthma and eosinophilia. ANCA is more common when GN is present.
Immune Complex Small-Vessel Vasculitis	
Vasculitis with moderate to marked vessel wall deposits of immunoglobulin and/or complement components predominantly affecting small vessels (capillaries, venules, arterioles, small arteries). GN is common.	
Anti-glomerular basement membrane (anti-GBM) disease	Vasculitis affecting glomerular capillaries, pulmonary capillaries, or both, with basement membrane deposition of anti-basement membrane autoantibodies. Lung involvement causes pulmonary hemorrhage, and renal involvement causes GN with necrosis and crescents.
Cryoglobulinemic vasculitis	Vasculitis with cryoglobulin immune deposits affecting small vessels (predominantly capillaries, venules, or arterioles) and associated with cryoglobulins in serum. Skin, glomeruli, and peripheral nerves are often involved.
IgA vasculitis (IgAV)	Vasculitis with IgA1-dominant immune deposits affecting small vessels (predominantly capillaries, venules, or arterioles). Often involves skin and gastrointestinal tract and frequently causes arthritis. GN indistinguishable from IgA nephropathy may occur.
Hypocomplementemic urticarial vasculitis (anti-C1q vasculitis)	Vasculitis accompanied by urticaria and hypocomplementemia affecting small vessels (capillaries, venules, arterioles) and associated with anti-C1q antibodies. GN, arthritis, obstructive pulmonary disease, and ocular inflammation are common.

Adopted by the 2012 Chapel Hill consensus conference on the nomenclature of systemic vasculitis. Note that all three categories affect arteries, but only small-vessel vasculitis has a predilection for vessels smaller than arteries.

ANCA, Antineutrophil cytoplasmic antibody; GN, glomerulonephritis; IgA, immunoglobulin A.

Modified from Charles P, Terrier B, Perrodeau E, et al. Comparison of individually tailored versus fixed-schedule rituximab regimen to maintain ANCA-associated vasculitis remission: Results of a multicentre, randomised controlled, phase III trial (MAINRITSAN2). *Ann Rheum Dis*. 2018;77:1143–1149.

Cutaneous involvement, manifesting as purpura, occurs frequently in all types of SVV (Fig. 26.4). Purpura is most common on the lower extremities, tends to occur as recurrent crops, and may be accompanied by areas of ulceration. Nodular cutaneous lesions are more frequent in GPA and EGPA than MPA. Nodules can be caused by dermal or subcutaneous arteritis and granulomatous inflammation.

Upper respiratory tract involvement is most common in GPA and EGPA but also occurs in MPA.²⁸ In all three categories, patients can have pulmonary hemorrhage caused by alveolar capillaritis. Patients with GPA, and to a lesser extent EGPA, can have pulmonary injury caused by necrotizing granulomatous inflammation, which may be detected radiographically as nodular or cavitating lesions. By definition, patients with MPA do not have granulomatous respiratory tract lesions.⁴

Manifestations of upper respiratory tract disease include subglottic stenosis, sinusitis, rhinitis, nasal septal collapse, otitis media, and ocular inflammation. The upper respiratory inflammation in MPA is caused by angiitis alone, without granulomatous inflammation. Destruction of bone, resulting in septal perforation and saddle nose, results from necrotizing granulomatous inflammation and does not occur in MPA.

Cardiac disease is identified in approximately 50% of patients with EGPA (usually myocarditis) but in less than 20% of patients with GPA or MPA. Cardiac manifestations in GPA and MPA are predominantly transient heart block and ventricular hypokinesis that respond to immunosuppressive treatment, although infarction, endocarditis, pericarditis, and severe life-threatening myocarditis may occur.

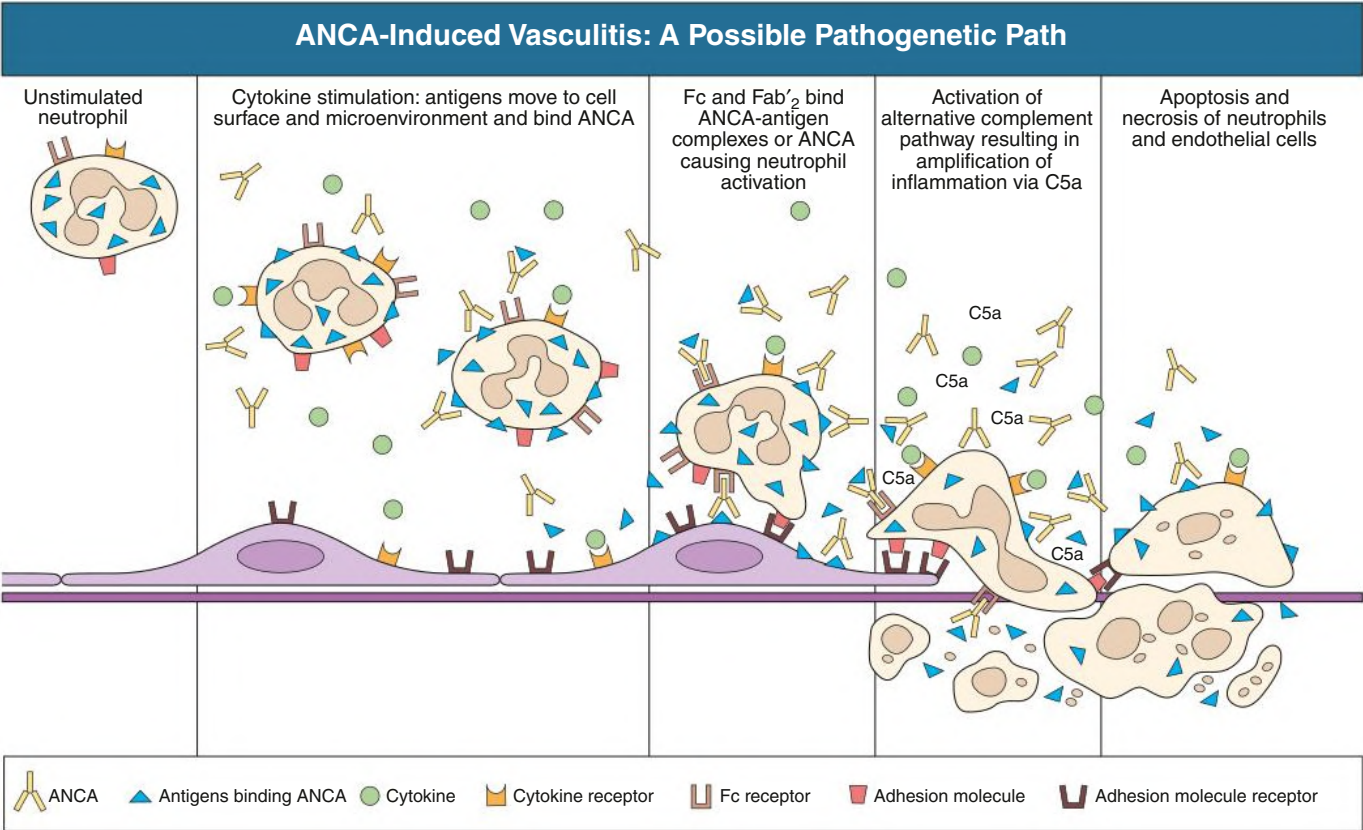


Fig. 26.3 Vasculitis Induced by Antineutrophil Cytoplasmic Antibody (ANCA). Hypothetical sequence of pathogenetic events.

TABLE 26.2 Organ System Involvement in Small-Vessel Vasculitis					
Organ System	FREQUENCY OF INVOLVEMENT (%)				
	Microscopic Polyangiitis	GPA	EGPA	IgA Vasculitis	Cryoglobulinemic Vasculitis
Kidney	90	80	20	50	55
Skin/cutaneous	40	40	40	90	90
Lungs	50	90	90	<5	<5
Ear, nose, throat	35	90	70	<5	<5
Musculoskeletal	60	60	50	75	70
Neurologic	30	50	40	10	40
Gastrointestinal	50	50	40	60	30

EGPA, Eosinophilic granulomatosis with polyangiitis; GPA, granulomatosis with polyangiitis; IgA, immunoglobulin A.

Peripheral neuropathy, usually mononeuritis multiplex, is the most common neurologic manifestation. Central nervous system involvement is less common, most often resulting from vasculitis within the meninges. Gastrointestinal (GI) involvement typically causes abdominal pain and hematochezia, with mesenteric ischemia and rarely intestinal perforation. Vasculitis in the pancreas and liver can mimic pancreatitis and hepatitis symptomatically and cause elevated serum pancreatic and liver enzymes.

Antineutrophil Cytoplasmic Autoantibody

Serologic testing for ANCA is useful for diagnosing SVV and pauci-immune crescentic GN but should be interpreted in the context of patient characteristics and performance qualities of the assay.^{29–32}

Antigen specificity of ANCA for PR3 versus MPO is helpful for diagnosis and has predictive value in clinical course and outcome.^{8,32} Laboratory testing for ANCA should include indirect immunofluorescence microscopy assay (IFA) and enzyme immunoassay (EIA).³¹ IFA using normal human neutrophils as substrate produces two major staining patterns (Fig. 26.5): cytoplasmic (c-ANCA), and perinuclear (p-ANCA). By EIA, most c-ANCAs have specificity for proteinase 3 (PR3-ANCA) and most p-ANCAs have specificity for myeloperoxidase (MPO-ANCA). For diagnostic accuracy, serologic testing for ANCA should include an immunochemical analysis for antigen specificity, such as an EIA.^{7,31,32} High-quality EIA can be used alone as a screening test for AAV.³² Although positive results are rare in healthy individuals, one-fourth of patients with other inflammatory kidney diseases



Fig. 26.4 Cutaneous Vasculitis. Ankle of a patient with small-vessel vasculitis, showing purpura and a few small ulcers.

TABLE 26.3 Antineutrophil Cytoplasmic Antibody in Small-Vessel Vasculitis

Disorder	FREQUENCY (%)		
	Proteinase 3 (PR3, Usually c-ANCA)	Myeloperoxidase (MPO, Usually p-ANCA)	Negative
Granulomatosis with polyangiitis	70	25	5
Microscopic polyangiitis	40	50	10
Eosinophilic granulomatosis with polyangiitis	5	40	55
Renal-limited pauci-immune crescentic GN	20	70	10

Approximate frequency of antibody in small-vessel vasculitis (ANCA) with specificity for proteinase 3 (PR3/c-ANCA) and for myeloperoxidase (MPO/p-ANCA) in patients with different categories of pauci-immune small-vessel vasculitis and crescentic glomerulonephritis.

GN, Glomerulonephritis.

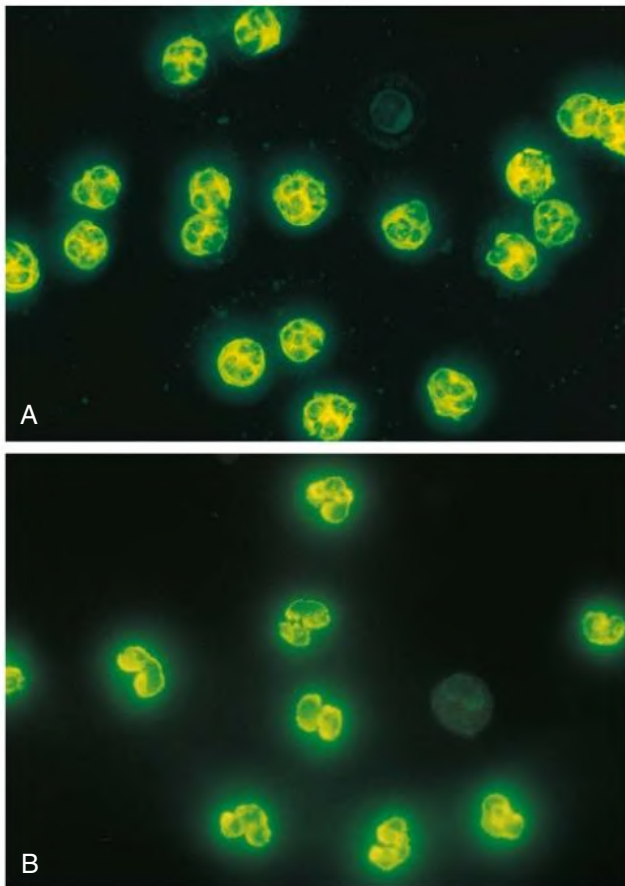


Fig. 26.5 Antineutrophil Cytoplasmic Antibodies (ANCAs). Indirect immunofluorescence staining pattern of alcohol-fixed normal human neutrophils. (A) Cytoplasmic pattern caused by ANCAs with specificity for proteinase 3. (B) Perinuclear pattern caused by ANCAs with specificity for myeloperoxidase (anti-immunoglobulin G). (Original magnification, $\times 250$.)

(especially lupus) have a false-positive IFA result (usually p-ANCA) and approximately 5% have a false-positive EIA result (usually low titer).³¹

ANCA testing has good sensitivity for AAV (80%–90%). The specificity and predictive value depend on the patient population and the assay quality.⁷ Although ANCAs are most frequent in patients with pauci-immune crescentic GN, one-fourth to one-third of patients with anti-glomerular basement membrane (GBM) crescentic GN and one-fourth of those with idiopathic immune complex crescentic GN with 50% or more crescents are ANCA-positive.^{27,33} Some patients have well-recognized types of immune complex GN complicated by ANCAs, such as membranous nephropathy and IgA nephropathy; others have non-lupus IgG-dominant immune complex disease that cannot be categorized further. Patients with concurrent ANCAs and anti-GBM antibodies have a worse prognosis than patients with ANCAs alone. This is complicated by antiperoxidase antibodies in some patients with anti-GBM exhibiting pulmonary-renal syndrome.³⁴ Peroxidase and MPO have antigenic overlap, which may be implicated in the autoimmune process. Anti-GBM antibodies typically disappear after treatment without recurrence, whereas ANCAs and associated vasculitis may recur.

Table 26.3 estimates the relative frequencies of PR3-ANCA/c-ANCA and MPO-ANCA/p-ANCA in the different clinical phenotypes of AAV, although this has geographic and racial influences.^{5,23} PR3-ANCA/c-ANCA are most prevalent in GPA, and MPO-ANCA/p-ANCA are most prevalent in renal-limited pauci-immune crescentic GN and EGPA. Patients with MPA have a more equal distribution of PR3-ANCA/c-ANCA and MPO-ANCA/p-ANCA, although this varies geographically.²³ Patients with EGPA have the lowest overall frequency of ANCAs, but the frequency of ANCAs is much higher in EGPA patients with GN (75%) than in those without GN (26%).³⁵ ANCA specificity correlates with clinical symptoms, with PR3-ANCA having the highest frequency (~90% when ANCA present) in patients who have destructive upper respiratory tract disease, especially saddle nose.²⁸

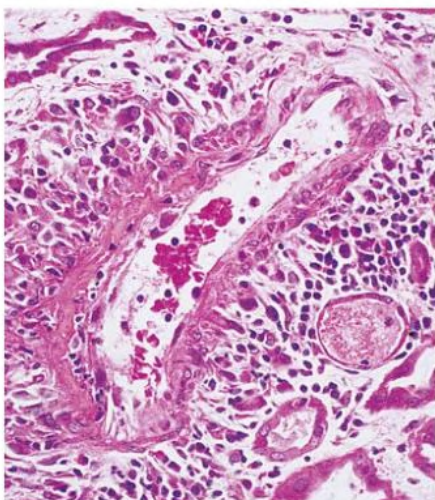


Fig. 26.6 Necrotizing Arteritis of Interlobular Artery in Patient with Antineutrophil Cytoplasmic Antibody–Associated Small-Vessel Vasculitis. There is segmental fibrinoid necrosis with adjacent perivascular leukocyte infiltration. (H&E, $\times 50$.)

Changes in ANCA titers over time may correlate with disease activity but are not dependable and must be interpreted with caution.^{7,31,32,36} Titers generally decrease with treatment and increase before or at disease recurrence. Increased ANCA titers should prompt careful patient evaluation for corroborating evidence of exacerbation, but most physicians do not modify treatment based on increased titer without accompanying clinical signs. Epitope-specific assays may provide better correlation with and prediction of disease outcome.¹⁰ MPO-ANCA with certain epitope specificities occur only in patients with active disease, disappear with remission, and reappear during relapse.¹⁰

From 10% to 20% of patients with pauci-immune necrotizing and crescentic GN and SVV are ANCA negative. The clinicopathologic and outcome characteristics of these patients are indistinguishable from ANCA-positive patients.³⁷ In the epitope-specific assays previously mentioned, some patients with AAV who are negative by clinical assays have MPO-ANCA with restricted epitope specificity that can be detected with sensitive assays.¹⁰

ANCAs may be positive but not associated with vasculitis in inflammatory conditions other than vasculitis, including inflammatory bowel disease (IBD), rheumatoid disease, chronic inflammatory liver disease, bacterial endocarditis, and cystic fibrosis. In IBD, specificity of the ANCAs usually is not against PR3 or MPO but against other neutrophil antigens, including lactoferrin, cathepsin G, and anti-bactericidal/permeability-increasing protein (BPI).^{31,32}

Pathology

The acute vascular lesion of pauci-immune SVV is segmental fibrinoid necrosis, often accompanied by leukocyte infiltration and leukocytoclasia.^{1,38–40} (Figs. 26.6 and 26.7). The earliest vasculitic lesions have infiltrating neutrophils quickly replaced by predominantly mononuclear leukocytes. The acute necrotizing lesions evolve into sclerotic lesions and may be complicated by thrombosis.

These focal necrotizing lesions can affect many different vessels, producing varied signs and symptoms. For example, involvement of glomerular capillaries causes nephritis; of alveolar capillaries, pulmonary hemorrhage; of dermal venules, purpura; of upper respiratory tract mucosal venules, rhinitis and sinusitis; of abdominal visceral arteries, abdominal pain; and of epineural arteries, mononeuritis multiplex.

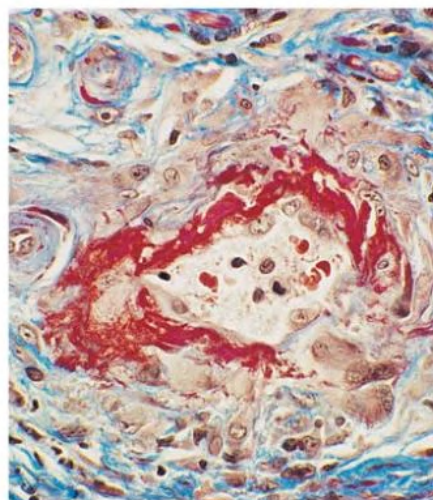


Fig. 26.7 Necrotizing Arteritis of Interlobular Artery in Patient with Antineutrophil Cytoplasmic Antibody–Associated Small-Vessel Vasculitis. The fibrinoid necrosis is indicated by the red staining. (Masson trichrome, $\times 100$.)

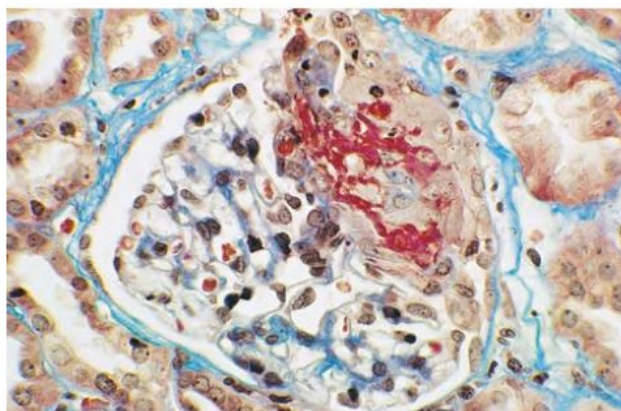


Fig. 26.8 Segmental Glomerular Necrosis and Crescent Formation in Patient with Antineutrophil Cytoplasmic Antibody–Associated Small-Vessel Vasculitis. The fibrinoid material is red. The uninvolved segments appear normal. (Masson trichrome, $\times 150$.)

Pauci-immune SVVs present as necrotizing GN, usually with resultant crescent formation.^{1,38–40} Early mild lesions have segmental fibrinoid necrosis with or without adjacent small crescent (Fig. 26.8). Severe acute lesions may have global necrosis with large circumferential crescents (Fig. 26.9). In a cohort of 181 kidney biopsy specimens from patients with ANCA-associated GN, 90% had glomerular crescents affecting 50% of glomeruli, with half having crescents in more than 50% of glomeruli.²⁷ Nonnecrotic segments within segmentally injured glomeruli (see Fig. 26.8) and glomeruli without necrosis have slight or no histologic abnormalities.

Approximately one-fourth of patients with anti-GBM crescentic GN and one-fourth of patients with immune complex–mediated crescentic GN are ANCA positive,²⁷ but less than 5% of patients with immune complex GN who do not have crescents are ANCA positive. Even in patients with immune complex GN, the presence of ANCAs is associated with an increased incidence of crescents (and inflammation in vessels other than glomerular capillaries). A histopathologic classification has been proposed for AAV GN: sclerotic class ($\geq 50\%$ globally sclerotic glomeruli), focal class ($\geq 50\%$ normal glomeruli), crescentic class ($\geq 50\%$ of glomeruli with cellular crescents), or mixed class if

none of these features predominated.⁴⁰ Kidney survival at 5 years was 93% for the focal class, 76% for crescentic class, 61% for mixed class, and 50% for sclerotic class.

In addition to GN, patients with AAV may have renal arteritis, most often affecting interlobular arteries (see Figs. 26.6 and 26.7), and medullary angiitis affecting the vasa recta (Fig. 26.10). Medullary angiitis rarely causes papillary necrosis.

Patients with GPA and EGPA have pathologic lesions in addition to necrotizing SVV.^{1,3,4} The necrotizing granulomatous inflammation

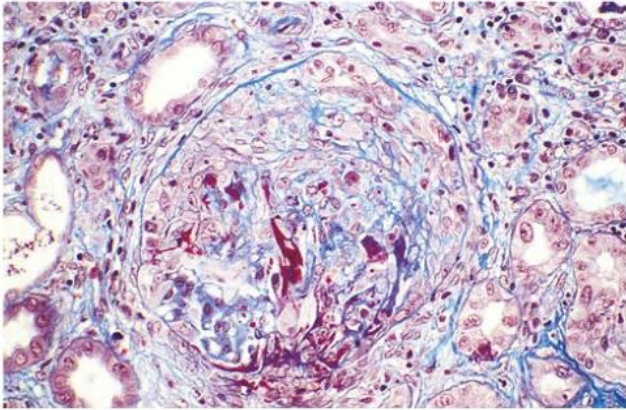


Fig. 26.9 Global glomerular necrosis and circumferential crescent formation in a glomerulus from patient with antineutrophil cytoplasmic antibody-associated small-vessel vasculitis. (Masson trichrome, $\times 150$.)

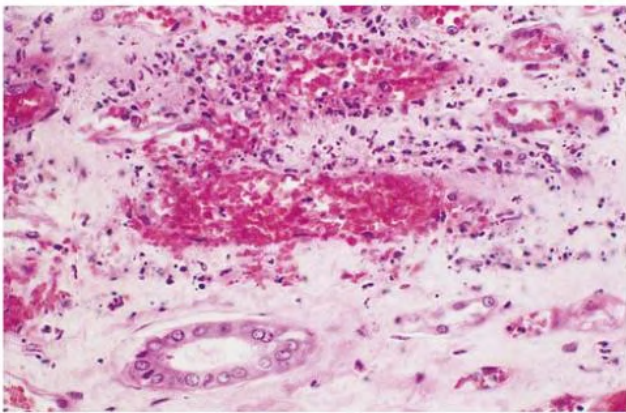


Fig. 26.10 Medullary leukocytoclastic angiitis involving vasa recta in a patient with granulomatosis with polyangiitis. (H&E, $\times 150$.)

of GPA occurs most often in the respiratory tract, characterized by necrotic zones surrounded by infiltrates of neutrophils, lymphocytes, monocytes, and macrophages, including scattered multinucleated giant cells. In patients with GPA and EGPA, extravascular granulomatous inflammation is rare in kidney biopsy specimens. Varying numbers of eosinophils may be present in the lesions of GPA, but these are more conspicuous in the necrotizing granulomatous inflammation of EGPA. Eosinophils are typically conspicuous in the vasculitic lesions of EGPA, but this is not a pathognomonic observation because eosinophils may be present in the vasculitic lesions of GPA, MPA, polyarteritis nodosa (PAN), and other vasculitides.

Differential Diagnosis

AAV must be differentiated from other forms of SVV that can produce similar signs and symptoms.^{1,3,4} An attempt should be made to distinguish MPA, GPA, and EGPA, although in some patients this cannot be accomplished conclusively and is not required for initiation of therapy. Patients should be categorized as MPO-ANCA, PR3-ANCA, or ANCA-negative because serotype has independent predictive value about clinical course and outcome.^{5,8,28} Pathologic confirmation of the granulomatous inflammation in ANCA disease is difficult because biopsy specimens often show nonspecific acute and chronic inflammation and necrosis. Findings such as nodular or cavitating lung lesions observed radiographically or destructive bone lesions in the nasal septum often are markers of necrotizing granulomatous inflammation to categorize patients. Because of treatment toxicity, even in patients with substantial clinical and serologic evidence of ANCA disease, pathologic confirmation of vasculitis is warranted. This can be accomplished with biopsy of different involved sites, including skin, muscle, nerve, gut, and kidney. In patients with substantial kidney involvement, kidney biopsy findings can be useful for predicting treatment response and clinical outcome.^{38–40}

All forms of SVV in Fig. 26.2 are capable of producing clinically indistinguishable overlapping features of disease, such as nephritis, purpura, peripheral neuropathy, myalgias, arthralgias, and abdominal pain. Table 26.4 lists features that help distinguish categories of SVV.³ Accurate differentiation is important for proper patient management because the natural histories and treatments vary greatly. For example, a patient with nephritis, arthralgias, and abdominal pain could have IgA vasculitis (IgAV, formerly Henoch-Schönlein purpura), MPA, cryoglobulinemic vasculitis, or several other small-vessel vasculitides. Serologic and pathologic observations are useful for reaching the correct diagnosis (see Table 26.4). A positive ANCA assay (confirmed by EIA to be MPO- or PR3-ANCA) supports a diagnosis of MPA or one of the other pauci-immune SVV. A negative ANCA assay and positive

TABLE 26.4 Differential Diagnostic Features of Select Forms of Small-Vessel Vasculitis

Features	Microscopic Polyangiitis	GPA	EGPA	IgA Vasculitis	Cryoglobulinemic Vasculitis
Vasculitic signs and symptoms ^a	+	+	+	+	+
IgA-dominant immune deposits	–	–	–	+	–
Cryoglobulins in blood and vessels	–	–	–	–	+
ANCA in blood	+	+	+	–	–
Necrotizing granulomas	–	+	+	–	–
Asthma and eosinophilia	–	–	+	–	–

^aThese vasculitides can manifest any of the shared features of small-vessel vasculitides, such as nephritis, purpura, abdominal pain, peripheral neuropathy, myalgias, and arthralgias. Each is distinguished by the presence and, just as important, by the absence of certain specific features.

ANCA, Antineutrophil cytoplasmic antibodies; EGPA, eosinophilic granulomatosis with polyangiitis; GPA, granulomatosis with polyangiitis.

Modified from Jennette JC, Falk RJ. Small-vessel vasculitis. *N Engl J Med*. 1997;337:1512–1523.

cryoglobulin assay (especially accompanied by hypocomplementemia and positive hepatitis C serology) support a diagnosis of cryoglobulinemic vasculitis. A negative ANCA assay, negative cryoglobulin assay, and normal complement levels support a diagnosis of IgAV, especially in a patient younger than 21 years. Patient age influences the likelihood of a specific diagnosis, so approximately 80% of children younger than 10 years who have purpura, nephritis, and arthralgias will have IgAV, but approximately 80% of adults older than 60 years with the same symptoms will have an AAV. However, each disease can occur at any age.

Drug exposures that may provoke AAV must be considered, including penicillamine, hydralazine, propylthiouracil,¹² and cocaine adulterated with levamisole.¹² Cholesterol embolization can mimic the clinical features of SVV with lower extremity rash or livedo reticularis, but ANCA assay is negative. The differential diagnosis of lung hemorrhage and nephritis includes anti-GBM disease, alone or with ANCA disease.

Natural History

Before the advent of immunosuppressive therapy, patient survival in MPA and GPA was poor, with most dying in less than 1 year. With adequate immunosuppressive therapy, 5-year kidney and patient survival is 65% to 75%.^{5,41,42} The success of long-term maintenance of kidney function is inversely correlated with the serum creatinine concentration when therapy begins, indicating the importance of early diagnosis and prompt therapeutic intervention. Patient survival increases with early treatment of pulmonary hemorrhage and sepsis and avoidance of overimmunosuppression leading to life-threatening infections. Adverse events from therapy, including infections, are the leading cause of death in the year after diagnosis.⁴¹ Older age, higher serum creatinine at presentation, pulmonary hemorrhage, and dialysis-dependent kidney failure correlate with an overall poor outcome; however, dialysis-dependent kidney failure may resolve with aggressive early therapy. Respiratory tract disease and PR3-ANCA are predictors of higher relapse rates.⁴² Pathologic features correlating with kidney outcome include histologically normal glomeruli, glomerular sclerosis, interstitial leukocyte infiltration, tubular necrosis, and tubular atrophy.⁴⁰ Increased histologically normal glomeruli or glomeruli with cellular crescents correlate with a better prognosis than higher proportions of globally sclerotic glomeruli, suggesting that active inflammatory lesions may be suppressed if not reversed by treatment, whereas chronic injury at initiation of treatment may be irreversible.

When severe GN is present, the kidney prognosis is similar for patients with MPA, GPA, or EGPA, and renal-limited pauci-immune crescentic GN. Kidney involvement is less common and usually less severe in patients with EGPA. Cardiac involvement is the most frequent cause of death in patients with EGPA but rarely causes mortality in MPA or GPA. GPA has a broad spectrum of clinical manifestations, from localized indolent disease to fulminant multisystem disease. Some patients have disease limited to the upper respiratory tract or the upper and lower respiratory tract. Limited disease may have a more benign natural history than systemic disease with substantial kidney involvement and may warrant less aggressive treatment.

Patients with MPO-ANCA have a slightly better kidney outcome than those with PR3-ANCA, although they have more kidney impairment and more chronic kidney pathologic changes at presentation. Patients with PR3-ANCA have more extrarenal organ manifestations (especially respiratory tract disease), more frequent relapse, and higher mortality than patients with MPO-ANCA. Regardless of the category of AAV, the best clinical predictor of kidney outcome is the glomerular filtration rate (GFR) at diagnosis.

Treatment

In patients with AAV affecting the kidneys, the goal is to avoid overtreating mild disease or undertreating severe disease. GN resulting in kidney impairment is an indication for immunosuppressive treatment in patients with GPA, MPA, EGPA, and renal-limited pauci-immune crescentic GN. Patients with AAV and concurrent immune complex disease should be treated similarly to patients with AAV alone. Patients with AAV and concurrent anti-GBM disease should be treated similarly to patients with anti-GBM disease alone. Treatment involves three phases: induction of remission, maintenance of remission, and treatment of relapse (Fig. 26.11).

Induction Therapy

Standard induction therapy for AAV combines corticosteroids with an immunomodulatory agent, usually cyclophosphamide or rituximab.^{5,42–55} Combined corticosteroid and cyclophosphamide induces remission in approximately 75% of patients at 3 months and 90% at 6 months. Combined induction regimens vary with respect to agents, doses, route of administration, and duration. One approach begins with methylprednisolone 7 mg/kg/day IV for 3 days, followed by oral prednisone 1 mg/kg/day, tapering to an alternate-day regimen and discontinuing within 3 to 6 months.⁴⁴ Alternatively, corticosteroids can be administered as prednisolone 1 mg/kg/day tapered to 0.25 mg/kg/day by 3 months.⁴⁸ Most trials used an IV methylprednisolone dose of 1 to 3 g followed by oral prednisone 1 mg/kg/day (max 80 mg).

Early steroid withdrawal is possible using combination therapy; 49 patients were successfully induced with a short course of oral glucocorticoids (1–2 weeks) while receiving combination cyclophosphamide (3 months) and rituximab (2 doses) therapy.⁵⁶ The Plasma Exchange and Glucocorticoids in Severe ANCA-Associated Vasculitis (PEXIVAS) trial demonstrated noninferiority of reduced corticosteroid dosing (60% less exposure) compared with standard dosing in end-stage kidney disease (ESKD) and all-cause mortality.⁵⁷ Novel complement-based strategies are being evaluated to reduce cumulative corticosteroid exposure. In a phase II trial, avacopan (CCX168; selective C5a receptor inhibitor) was as effective as prednisone in patients with newly diagnosed or relapsing vasculitis.⁵⁸ The phase III Avacopan Development in Vasculitis to Obtain Corticosteroid Elimination and Therapeutic Efficacy (ADVOCATE) trial for safety and efficacy of avacopan compared with prednisone showed that 72.3% of patients achieved remission with avacopan compared with 70.1% with prednisone.^{59,60} Other complement-based trials are underway with IFX-1, an anti-C5a antibody (NCT03712345, NCT03895801).

Because of the morbidity with long-term corticosteroid use, a reduced-dose corticosteroid regimen and discontinuation by 16 weeks in most patients is recommended. Utilization of cyclophosphamide or combination of cyclophosphamide and rituximab can help taper steroids faster.

Corticosteroid treatment is combined with oral cyclophosphamide 2 mg/kg/day or cyclophosphamide at 0.35 to 0.5 g/m² IV per month adjusted upward to 1 g/m², based on the leukocyte count after 2 weeks, with a target nadir of 3000 cells/mm³. Oral cyclophosphamide dose can be reduced by 25 mg for patients older than 60 years. A comparison by the European Vasculitis Study Group (CYCLOPS trial) indicated that IV cyclophosphamide has similar remission rates as oral cyclophosphamide while reducing the total cyclophosphamide dose.⁵²

The US Food and Drug Administration (FDA) and European Medicines Agency (EMA) approved rituximab for AAV induction therapy based on the results of two randomized controlled trials (RCTs), noting that rituximab-based cyclophosphamide-sparing strategies were comparable to traditional cyclophosphamide induction protocols.^{50,51} The Rituximab for ANCA-Associated Vasculitis (RAVE) trial⁵¹ noted no difference in remission off all therapy in 6

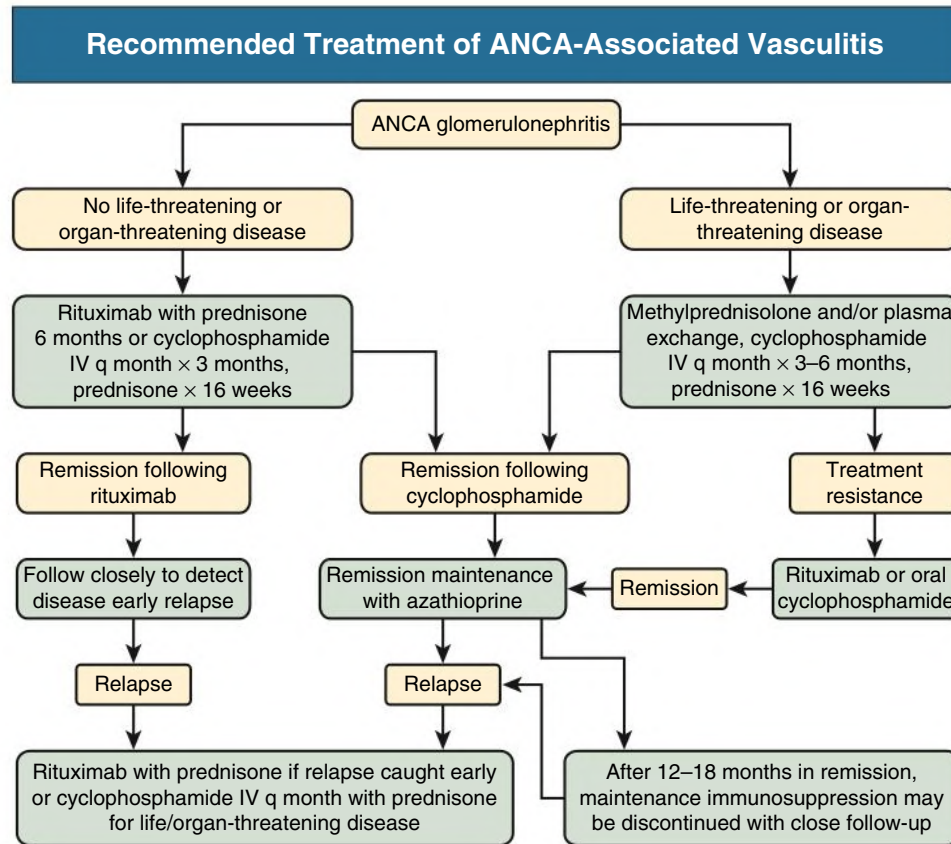


Fig. 26.11 Recommended treatment for antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis.

months between patients treated with prednisone and four infusions of 375 mg/m² rituximab versus oral cyclophosphamide (64% in rituximab arm, 55% in cyclophosphamide arm). The RITUXVAS trial^{61,62} compared 6 to 10 infusions of cyclophosphamide followed by azathioprine to 4 infusions of 375 mg/m² rituximab in combination with 2 infusions of cyclophosphamide without maintenance therapy. There was no difference in remission rates at 12 months between the groups. In RAVE and RITUXVAS, there were no significant differences in adverse events between rituximab and cyclophosphamide. Compared with cyclophosphamide, rituximab is more effective in PR3-ANCA disease.⁶³ Our approach uses two 1-g doses 14 days apart.

Combination of cyclophosphamide-rituximab allows a rapid steroid taper and reduces mortality, relapses, and ESKD compared with propensity-matched patients from European Vasculitis Study Group (EUVAS) trials.⁶⁴ We recommend combination IV cyclophosphamide-rituximab in patients with severe kidney or pulmonary involvement to decrease cumulative cyclophosphamide exposure.

Plasma exchange (PLEX) has come under much scrutiny. Prior studies showed benefit in patients with life-threatening pulmonary hemorrhage⁵³ and patients who have dialysis-dependent kidney failure at presentation.⁵⁴ With pulmonary hemorrhage, 20 of 20 patients treated with early PLEX had resolution of pulmonary bleeding compared with a 50% mortality in historical controls.⁵³ A EUVAS trial (MEPEX) evaluated the efficacy of IV methylprednisolone as induction therapy or PLEX in patients who had a serum creatinine of more than 500 μmol/L.⁵⁴ Patient survival and adverse events were similar in patients who did or did not receive PLEX. PLEX was associated with a 24% reduced risk for progression to ESKD, from 43% to 19% at 1 year.⁵⁴ Conversely, the PEXIVAS trial noted no reduction in mortality or ESKD with the use of PLEX in patients with severe AAV. The PEXIVAS trial enrolled 704 patients with severe AAV (an eGFR rate

of <50 mL/min/1.73 m² or diffuse pulmonary hemorrhage). ESKD or death occurred in 28.4% patient in the PLEX group versus 31% patients in control group.⁵⁷ One drawback of PEXIVAS was the lack of kidney biopsy data on interstitial fibrosis and tubular atrophy. PLEX had a trend for benefit in subgroup analyses of patients with pulmonary hemorrhage. Hence, we still consider PLEX in patients with acute kidney injury (estimated GFR [eGFR] < 30 mL/min/1.73 m²) without significant chronic changes of interstitial fibrosis. We also use PLEX in patients presenting with severe pulmonary hemorrhage requiring mechanical ventilation or supplemental oxygen.

Maintenance Therapy

Duration of induction therapy and intensity of maintenance therapy should be decreased as much as possible to reduce toxic side effects. This is difficult because of the tendency for AAV to recur. Various approaches to reduce the cyclophosphamide dose include using IV cyclophosphamide rather than oral, substituting a less toxic maintenance drug after 3 to 6 months, and discontinuing therapy earlier in patients with lower risk for relapse. IV cyclophosphamide regimens afford one-third to one-half the total dose of cyclophosphamide given in oral regimens. Long-term follow-up notes longer time to relapse in the oral cyclophosphamide treated group, at the cost of increased leukopenia.⁵⁵ Azathioprine after 3 to 6 months of therapy can reduce cyclophosphamide dose.⁴⁸ In another EUVAS trial (CYCAZAREM), cyclophosphamide was replaced with azathioprine 2 mg/kg/day after 3 to 6 months with no change in the relapse rate at the end of the study.⁴⁸ When azathioprine 2 mg/kg/day was compared with mycophenolate mofetil (MMF) 2 g/day in a randomized control fashion, relapses were more common in the MMF group compared with the azathioprine group (unadjusted hazard ratio [HR] for MMF, 1.69; 95% confidence interval [CI] 1.06–2.70). Adverse events, disease activity score, GFR,

and proteinuria did not differ between groups, but two instances of bladder cancer and three of skin cancer were noted in the azathioprine group compared with one instance of skin cancer in MMF group (HR, 0.25; 95% CI, 0.02–2.62).⁶⁵

The French Vasculitis Study Group compared methotrexate with azathioprine as maintenance therapy in patients with GPA and MPA.⁶⁶ Methotrexate was as effective as azathioprine for maintenance of remission, but methotrexate did not have fewer side effects. Methotrexate should not be used in patients with eGFR less than 30 mL/min/1.73 m². Karras et al. compared 24 versus 48 months of azathioprine remission maintenance therapy. Although 48 months of azathioprine therapy had fewer relapses (22% vs. 63%), more severe adverse events were noted (9 vs. 3 events).⁶⁷ An alternative is to stop all immunosuppressive therapy at 6 to 12 months if the patient is in full remission, especially if the patient is at lower risk for relapse.

Rituximab is another option for maintenance of remission.^{66,67} The MAINRITSAN trial compared rituximab with azathioprine for maintenance in patients with MPA and GPA in remission after induction treatment with cyclophosphamide and glucocorticoids. Rituximab was more effective than azathioprine for preventing relapse, including renal relapse,⁶⁶ for up to 60 months.⁶⁸ The rituximab versus azathioprine in remission maintenance (RITAZAREM) trial is evaluating patients with relapsing ANCA vasculitis after achieving remission with rituximab induction. Preliminary results suggest a lower relapse rate with rituximab as opposed to azathioprine (18% vs. 38%).^{69,70}

Dose and frequency of rituximab infusion and duration of therapy have been controversial. Tailored rituximab dosing of 500 mg based on reappearance of CD19⁺B cells or ANCA or rise in ANCA titer had similar relapse rates compared with fixed 500 mg rituximab dose every 6 months in the MAINRITSAN2 study (17.3% vs. 9.9%, $P = .22$).⁷¹ The tailored group had fewer infusions of rituximab (median 3 vs. 5), suggesting that dose and frequency of rituximab infusion can be successfully modified. The MAINRITSAN3 trial treated patients in remission from MAINRITSAN2 with placebo versus rituximab 500 mg biannually. The rituximab group had more major relapse-free survival compared with placebo group (100% vs. 87%, $P = .009$).⁷² Although the adverse events were similar, the trial included patients who had previously tolerated rituximab.

Because rituximab is a biologic agent that has immunogenic mouse protein in its structure, many patients have become sensitized and some have developed serum sickness–like disease. If sensitivity develops during maintenance therapy, rituximab may not be an option for recurrent clinical disease. Fully humanized anti-CD20 antibodies, ofatumumab, obinutuzumab, and ocrelizumab may have a lower risk of sensitization but efficacy data is lacking. The expense of rituximab may be prohibitive for some patients. Because of these concerns and the risk of hypogammaglobulinemia, we suggest using a rituximab maintenance dose of 500 mg. Additionally, because of ease of reinduction,⁷³ we recommend discontinuing rituximab if patients are in long-term remission (>2 years).

The role of antimicrobial agents such as trimethoprim-sulfamethoxazole (TMP-SMX) in maintenance of remission is controversial. Some studies suggest a benefit, but others do not because of an increased likelihood of relapse.⁷⁴ TMP-SMX may be useful adjunct therapy, especially in patients with upper respiratory tract disease but should not be used in the absence of immunosuppressive drugs with more proven efficacy (e.g., cyclophosphamide, azathioprine) for induction or maintenance therapy for systemic vasculitis or GN. We often use TMP-SMX as prophylaxis in patients receiving prednisone at 20 mg or higher dose.

Relapse Therapy

One-fourth to one-half of patients with AAV will experience a relapse within several years. Relapses are diagnosed by clinical and pathologic

evidence of recurrent disease, not by increase in ANCA titer alone.³⁶ Increase in ANCA titer does increase the likelihood of a relapse, and some advocate preemptive immunosuppressive therapy if the titer increases by at least fourfold.³⁶ We prefer identifying clear-cut clinical or pathologic evidence of relapse before increasing or resuming immunosuppressive therapy.⁴⁴

Observational data suggest that rituximab is the best treatment for relapses and is superior to cyclophosphamide.⁵⁰ Reinstitution of treatment similar to an induction regimen is used most often, but less intensive or less toxic therapy may be adequate.⁴⁴ Lifetime exposure to cyclophosphamide is a consideration when selecting treatment for relapses. In addition to rituximab or cyclophosphamide, other therapies can be used for treating relapse, including azathioprine, MMF, methotrexate, or combinations of these drugs tailored to the individual patient with recalcitrant disease.

The MAINRITSAN and RITAZAREM trials suggest rituximab may be more effective in reducing relapse rates compared with azathioprine.^{66,69,70} Long-term therapy may not be required, especially when there is a low risk for relapse. In our Glomerular Disease Collaborative Network (GDCN) cohort, 277 patients (65%) of 427 patients stopped immunosuppressive therapy. Some patients (23%) stayed off therapy for 5 or more years, suggesting long-term remission off therapy is possible. Additionally, 197 patients never relapsed with no difference between those who were on therapy or off.⁷³ In prospective observational studies from the GDCN, the relative risk for relapse was increased in individuals who had PR3-ANCA and respiratory tract disease.⁴² Patients had more than a threefold risk for relapse compared with individuals having MPO-ANCA and without lung or ear, nose, and throat disease. This model of relapsing disease was investigated in a separate registry in France in which the combination of PR3-ANCA and lung disease was the most important predictive marker.⁷⁵

Comparison of the major rituximab-based trials (RAVE, RITUXVAS, MAINRITSAN1, MAINRITSAN2, and RITAZAREM) shows a significant underrepresentation of MPO-ANCA patients compared with PR3-ANCA. There were 2 to 3 times as many PR3-ANCA patients compared with MPO-ANCA patients in the major rituximab-based trials. Because PR3-ANCA predominates in these trials (232 vs. 111), it is likely that the number of flares seen was higher than what would have been expected with MPO-ANCA patients. Thus, in patients having a much smaller risk for relapse, all therapy may be stopped, provided the patient is closely monitored (e.g., home urinary dipstick testing to monitor for recurrence of hematuria). The risk for relapse is about 10% to 15% even with maintenance therapy.

Transplantation

Kidney transplantation is not contraindicated in patients with ESKD caused by AAV. In a multicenter experience, the vasculitis relapse rate was 0.02 per patient per year⁷⁶ (see [Chapter 113](#)). Although a positive ANCA titer at the time of transplantation is not thought to increase the risk for recurrent disease in the transplant,⁷⁷ a study of 16 patients suggested a higher likelihood of relapse with PR3-ANCA compared with MPO-ANCA (OR, 2.19; $P = .71$).⁷⁸ An increase in ANCA titer and an active urine sediment suggest recurrent GN, but the diagnosis requires pathologic confirmation. Recurrent ANCA GN in a kidney transplant responds similarly to recurrent disease in native kidneys and usually is treated with cyclophosphamide and corticosteroids. Rituximab has been used successfully and avoids the risk for cyclophosphamide toxicity.⁷⁹

POLYARTERITIS NODOSA

PAN is a systemic necrotizing arteritis that affects predominantly main visceral arteries and their intraparenchymal branches.^{2,26,80} The Chapel

Hill nomenclature system limits the diagnosis of PAN to patients who have only arteritis.⁴ The presence of vasculitis in capillaries and venules excludes a diagnosis of PAN and indicates SVV. Thus, GN excludes a diagnosis of PAN. When PAN is distinguished from MPA, the two categories of vasculitis have different pathologic characteristics, clinical features, and natural histories.⁸¹

Pathogenesis

The etiology and pathogenesis of PAN are unknown and probably are diverse.² When PAN is separated from MPA, the latter is associated with ANCA. An immune complex trigger for PAN has been proposed but not confirmed. A minority of patients have hepatitis B virus (HBV) infection, suggesting that the HBV infection is producing immune complexes that localize in arterial walls and induce inflammation, ultimately leading to narrowing and weakening of vessel walls, thereby predisposing to aneurysms.² However, evidence that HBV infection is causing vascular immune complex deposition is stronger in certain forms of GN and SVV than in PAN.

Epidemiology

When defined by the Chapel Hill nomenclature system, PAN has a low prevalence of 1.5 per million.²³ Prevalence is higher in regions with higher levels of endemic HBV infection. PAN affects males and females equally and is found in all races. Onset occurs most frequently between ages 40 and 60 years.

Clinical Manifestations

The usual clinical presentation of PAN includes nonspecific constitutional symptoms: fever, malaise, arthralgias, myalgias, weight loss, and manifestations of arteritis.^{39,81,82} Peripheral neuropathy, typically in the form of mononeuritis multiplex, is common. This is caused by inflammation of small epineural arteries and is clinically indistinguishable from the peripheral neuropathy caused by other forms of vasculitis affecting epineural arteries, such as MPA, GPA, and EGPA. GI involvement occurs in about half of patients, usually manifesting as abdominal pain and hematochezia. Bowel infarction is uncommon and perforation is rare. Kidney involvement produces infarction and hemorrhage, as indicated by flank pain and hematuria. Rupture of an arterial aneurysm with retroperitoneal or peritoneal hemorrhage is an uncommon but potentially lethal renal complication. Approximately one-third of patients develop hypertension, rarely reaching the malignant range. Red, tender inflammatory nodules are the most common cutaneous manifestation. Infarction, ulceration, and livedo reticularis may be present.

Arterial aneurysms may be detected by angiography in patients with PAN (Fig. 26.12). This is not a completely specific determination because any necrotizing arteritis that affects arteries large enough to be seen by angiography can produce this finding.

Pathology

Any artery in the kidney can be affected by PAN, although the interlobar and arcuate arteries are affected most often.¹⁷ Nodular inflammatory lesions and aneurysms (pseudoaneurysms) can be observed grossly when medium-sized arteries are involved. Inflammation in small arteries can be observed only by microscopy.

The characteristic acute lesion is segmental transmural fibrinoid necrosis of arteries, usually accompanied by infiltrating leukocytes with leukocytoclasia³ (Fig. 26.13). The earliest lesions have numerous neutrophils, and later lesions have predominantly mononuclear leukocytes. Acute lesions may be complicated by thrombosis or hemorrhage. Older lesions develop fibrosis and end-arterial remodeling. The aneurysms of necrotizing arteritis are not true aneurysms but

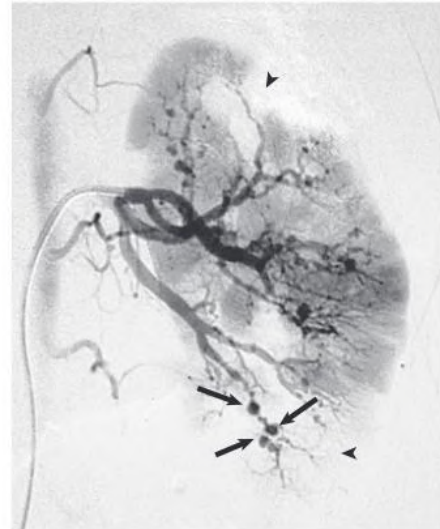


Fig. 26.12 Renal Angiogram in Polyarteritis Nodosa. Angiogram shows patchy kidney perfusion defects (arrowheads) and aneurysms (arrows).

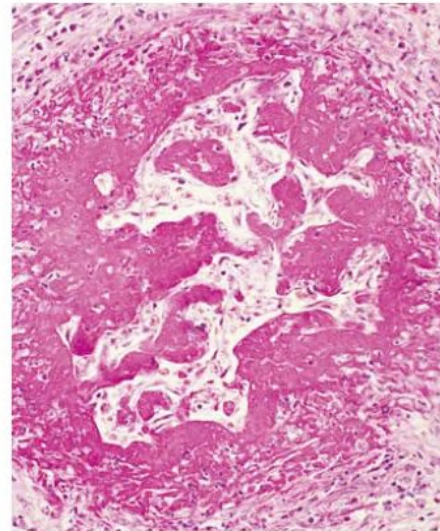


Fig. 26.13 Necrotizing Arteritis in Arcuate Artery of Patient With Polyarteritis Nodosa. The lumen is partially occluded by thrombotic material that is continuous with the fibrinoid material that has replaced the entire wall of the artery. (H&E, x50.)

rather inflammatory pseudoaneurysms—the walls of the arteries are not dilated but have been eaten away by the necrotizing inflammation, which then erodes into the surrounding perivascular tissue to create an enlarged lumen. This explains the propensity for such lesions to induce thrombosis or rupture.

The necrotizing arteritis of PAN is indistinguishable by light microscopy from arteritis caused by other necrotizing vasculitides affecting arteries.^{2,3} For example, necrotizing arteritis in a skeletal muscle biopsy specimen or a peripheral nerve biopsy specimen is histologically identical whether caused by PAN, MPA, GPA, or EGPA. For these vasculitides to be distinguished, additional clinical and serologic information is required.

Differential Diagnosis

Clinical features may assist in distinguishing PAN from other forms of vasculitis, especially other forms of necrotizing vasculitis that can affect arteries, such as MPA^{2,3,83} (Table 26.5). A positive ANCA test supports

TABLE 26.5 Clinical Differences Between Polyarteritis Nodosa and Microscopic Polyangiitis

Clinical Feature	Polyarteritis Nodosa	Microscopic Polyangiitis
Microaneurysms by angiography	Yes	No (rare)
Rapidly progressive nephritis	No	Yes (very common)
Pulmonary hemorrhage	No	Yes
Renovascular hypertension	Yes (10%–33%)	No
Peripheral neuropathy	Yes (50%–80%)	Yes (10%–20%)
Positive hepatitis B serology	Uncommon	No
Positive ANCA	Rare	Frequent
Relapses	Rare	Frequent

ANCA, Antineutrophil cytoplasmic antibody.

Modified from Scott DGI, Watts RA. Epidemiology and clinical features of systemic vasculitis. *Clin Exp Nephrol*. 2013;17:607–610.

the diagnosis of one of the AAVs rather than PAN. The presence of GN indicates some form of SVV rather than PAN. Vasculitic pulmonary disease is rare in PAN but common in MPA, GPA, and EGPA. Peripheral neuropathy or muscle tenderness with arteritis in epineural or skeletal muscle arteries is not a useful differentiating feature because it often occurs in PAN and in AAV. Kawasaki disease (KD) causes necrotizing arteritis but is distinguished from PAN by the presence of the mucocutaneous lymph node syndrome.

Natural History

The natural history of PAN is muddled because early studies grouped MPA with PAN. PAN with multisystem involvement has a poor prognosis without therapy.²⁶ The 10-year survival with appropriate treatment is approximately 80%. Approximately 15% of patients who enter remission do relapse, which is much less frequent than with MPA. Relapse is more likely if treatment is delayed.

Treatment

PAN in patients without HBV infection is treated with corticosteroids and cytotoxic drugs (usually cyclophosphamide) if there is life-threatening major organ involvement.^{26,81,82} Regimens vary and include treatments similar to those described earlier for MPA and GPA. In patients with no risk factors for poor outcome (e.g., age > 50 years; cardiac, gut, or kidney involvement), corticosteroids alone may be adequate and are less toxic than corticosteroids combined with cytotoxic agents.⁸² In patients without major organ involvement, azathioprine or methotrexate can be used to reduce steroid exposure.

Aggressive immunosuppressive therapy without initial antiviral therapy is contraindicated in patients with HBV-associated PAN because of potential adverse effects on the outcome of the HBV infection. Patients with hepatitis B viremia should be treated with effective hepatitis B therapy. Often additional immunosuppression is needed, but short-term corticosteroid treatment combined with antiviral agents and possibly PLEX should precede more extensive immunosuppression in such patients.

KAWASAKI DISEASE

Definition

KD is an acute febrile illness that usually occurs in young children, often under 1 year, and is characterized by the mucocutaneous lymph node (MCLN) syndrome (see later discussion).^{2,4,84–86} Necrotizing

arteritis is a complication of KD that is present in some patients. Clinically significant kidney involvement is rare and KD is seldom encountered by nephrologists. If untreated, approximately 20% to 25% of children develop coronary artery aneurysms that may cause myocardial infarction.⁸⁵ Most patients undergoing autopsy after fatal myocardial infarction (MI) caused by coronary artery arteritis also have vasculitis in renal arteries.⁸⁴

Pathogenesis

The occasional occurrence of KD as an endemic or epidemic disease suggests that the cause may be an infectious agent or environmental toxin.² Both cell-mediated and antibody-mediated mechanisms have been incriminated, as well as dysregulated innate immunity, possibly mediated by innate immune pathogen-associated molecular patterns.⁸⁶ At present, the etiology and pathogenesis of KD are unknown.

Epidemiology

KD usually occurs in children younger than 5 years (median, 2–3 years).⁸⁵ First described in Japan, it occurs worldwide but is more common in Asians and Polynesians than in Whites and Blacks. In Japan, the incidence is 50 in 100,000 children younger than 5 years, with 50% of the children younger than 2 years.^{23,85} KD occasionally occurs in an endemic or epidemic pattern but usually is sporadic (with the COVID-19 pandemic, there have been cases of a Kawasaki-like disease in children after COVID-19 infection). KD is not transmitted person to person and does not occur in clusters within households, schools, or nurseries.⁸⁶

Clinical Manifestations

MCLN syndrome is the characteristic clinical manifestation of KD.^{4,85,86} This includes fever (temperature of 38°C to 40°C), mucosal inflammation, swollen red tongue (strawberry tongue), polymorphous erythematous rash, indurative edema of the extremities, erythema of palms and soles, desquamation from the tips of digits, conjunctival injection, and enlarged lymph nodes. Frequency of active arteritic lesions peaks during the first week of the illness and greatly reduces after 1 month. Arteritis often manifests as cardiac disease. Thrombosis of inflamed coronary arteries in patients with KD is the most common cause of childhood MI. Clinically significant kidney disease is surprisingly uncommon given that autopsy reveals arteritis in kidney vessels in up to three-fourths of patients.⁸⁴

Pathology

KD involves small- and medium-sized arteries. The acute histologic lesion is necrotizing inflammation with less fibrinoid necrosis and more vessel wall edema than usually observed with PAN⁸⁴ (Fig. 26.14). Aneurysm (pseudoaneurysm) formation and thrombosis may occur. The most frequent site of arteritis is the coronary arteries, followed by the kidney arteries.⁸⁴ Arteritis most often affects interlobar arteries, occasionally arcuate arteries, and rarely interlobular arteries.

Differential Diagnosis

KD is sometimes misdiagnosed as childhood PAN. The differentiation of KD from PAN is important because corticosteroid treatment may increase the risk for coronary artery aneurysms in KD. Arteritis in a child younger than 5 years should always raise the possibility of KD. The presence or absence of the MCLN syndrome is the basis for distinguishing between KD and other forms of arteritis.⁴

Natural History

KD usually is self-limited, with an uneventful recovery if treated promptly with IV gamma globulins.^{84,85} Recurrence occurs in less than 5% of patients.⁸⁵

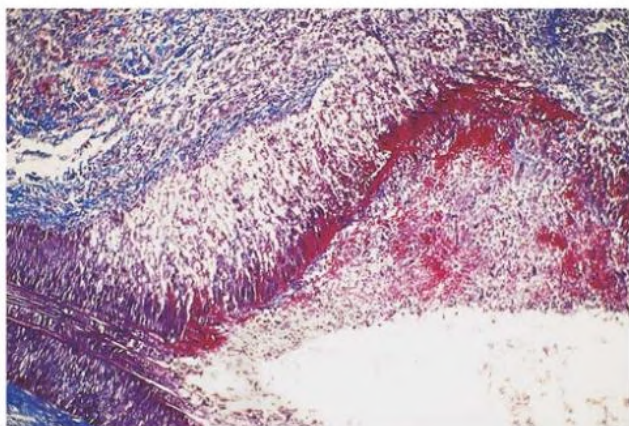


Fig. 26.14 Kawasaki Disease Arteritis Affecting Renal Interlobar Artery in a Young Child. The artery wall is intact on the far left. The remainder of the wall has extensive edema, infiltration by mononuclear leukocytes, and a band of fuchsinophilic (red) fibrinoid material roughly at the junction between the inflamed intima and muscularis. (Masson trichrome, $\times 25$.)

Treatment

Aspirin and single IV immunoglobulin infusion (2 g/kg) within 10 days of fever onset are the standard therapy for KD.^{84,85} Moderate to high doses of aspirin are used until resolution of fever followed by low-dose for antiplatelet effect. The role of corticosteroid treatment is controversial but may be beneficial in patients with severe disease that does not respond well to aspirin and intravenous immunoglobulin.

TAKAYASU ARTERITIS AND GIANT CELL ARTERITIS

Takayasu arteritis and giant cell arteritis (GCA) affect the aorta and its major branches more often than other forms of vasculitis.^{4,87,88} GCA has a predilection for the extracranial branches of the carotid artery but can affect arteries in almost any organ. Takayasu arteritis has a predilection for major arteries supplying the extremities. Both diseases cause chronic vascular inflammation, often with a granulomatous appearance that may include multinucleated giant cells. GCA, but not Takayasu arteritis, is associated with polymyalgia rheumatica (PMR). The relation of Takayasu arteritis and GCA is not known. The striking demographic differences suggest that they are distinct pathophysiologic entities.

Pathogenesis

The etiology and pathogenesis of GCA and Takayasu arteritis are unknown.² Because of the histologic changes and the nature of the infiltrating leukocytes, cell-mediated immune mechanisms are incriminated. The inciting antigen or autoantigen has not been identified.

Epidemiology

Takayasu arteritis is seen most frequently in Asia. GCA occurs most often in individuals of northern European ancestry. Takayasu arteritis has a female-to-male ratio of approximately 9:1 and GCA of 4:1. Takayasu arteritis usually is diagnosed in those between ages 10 and 20 years and is rare after age 50 years. GCA is rare before age 50 years. Takayasu arteritis has uniform global incidence of 1 to 2 per million.²³ Race and ethnicity affect the vascular distribution of Takayasu arteritis, with the aortic arch and its branches affected mainly in Japanese patients, abdominal aorta and its branches affected mainly in Indian patients, lower abdominal aorta affected mainly in African

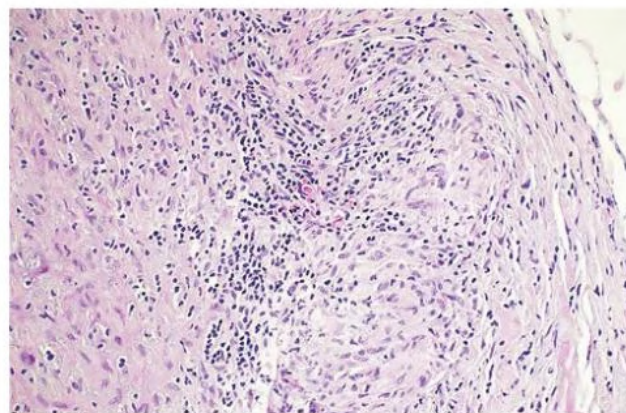


Fig. 26.15 Severe Giant Cell Arteritis Affecting a Main Renal Artery. This caused marked renal atrophy and renovascular hypertension. (H&E, $\times 50$.)

populations, and kidney involvement most common in Asian and African populations.⁸⁸

GCA is the most common form of vasculitis, with an incidence highest in populations of Scandinavian descent, with an annual incidence of 15 to 35 per 100,000 older than 50 years.²³

Clinical Manifestations

In addition to constitutional symptoms, such as fever, arthralgias, and weight loss, the major clinical manifestations of Takayasu arteritis and GCA are caused by arterial narrowing and resultant ischemia.^{2,87} Major clinical manifestations of Takayasu arteritis are reduced pulses (95% of patients), vascular bruits, claudication, and renovascular hypertension. Renovascular hypertension results from kidney ischemia caused by renal artery stenosis or aortic coarctation.⁸⁹ Reduced aortic elasticity and impairment of carotid artery baroreceptors also may play a role in some patients. The European League Against Rheumatism (EULAR) recommends thorough imaging assessment (see later discussion) of the entire major arterial tree when a diagnosis of Takayasu arteritis is suspected.⁹⁰

Headache is the most common presenting symptom in patients with GCA. Temporal artery tenderness, nodularity, or decreased pulsation is present in half of patients. Other symptoms include blindness, deafness, jaw claudication, tongue dysfunction, extremity claudication, and reduced pulses. More than half of patients with GCA have PMR, characterized by stiffness and aching in the neck and the proximal muscles of the shoulders and hips. Clinically significant kidney disease is less common in GCA than in Takayasu arteritis. Case reports of necrotizing and crescentic GN associated with GCA may represent examples of GPA or MPA with temporal artery involvement.

Computed tomography (CT), magnetic resonance angiography (MRA), fluorodeoxyglucose positron emission tomography/CT (FDG-PET/CT), and contrast-enhanced ultrasonography are used to diagnose and assess the activity of vascular inflammation in Takayasu arteritis and GCA.⁸⁸ CT and MRA identify structural changes (e.g., stenosis, aneurysms). FDG-PET/CT detects active inflammation even in vessels that show no overt structural changes by other imaging techniques.

Pathology

The aortitis and arteritis of Takayasu arteritis and GCA cannot be confidently differentiated by pathologic examination.⁴ Both are characterized in the active phase by inflammation with a predominance of mononuclear leukocytes, often with scattered multinucleated giant cells (Fig. 26.15). The chronic phase is characterized by progressive

fibrosis that may cause severe narrowing of vessels, with resultant ischemia. Major renal arteries are often found to be involved at autopsy in patients with Takayasu arteritis and those with GCA. However, clinically significant kidney disease is relatively common in Takayasu arteritis but rare in GCA. A glomerular lesion characterized by nodular mesangial matrix expansion and mesangiolysis may occasionally be a component of Takayasu arteritis.⁹¹

Differential Diagnosis

There is overlap between the clinical manifestations and pathologic features of Takayasu arteritis and GCA. Patient age and presence or absence of PMR are the best factors for distinguishing between these vasculitides.⁴ GCA has been called temporal arteritis, but not all patients have temporal artery involvement, and patients with other types of vasculitis (PAN, GPA, MPA) can have involvement of the temporal arteries.

Treatment

Corticosteroids are the usual treatment of GCA and Takayasu arteritis.⁹⁰ EULAR recommends initial daily therapy with prednisolone 1 mg/kg until disease control followed by tapering over several months to over a year.^{90,92} Prolonged treatment may be dictated by persistent disease activity. Rapid steroid taper is associated with increased risk of relapse.⁹³ Tocilizumab, an interleukin (IL)-6 inhibitor, has been a successful adjunct to corticosteroids in patients at high risk for corticosteroid-related adverse events or with life- or organ-threatening disease.^{92–94} Methotrexate is an alternative. Cytotoxic agents such as cyclophosphamide may be required in patients with recalcitrant disease. Patients with GCA are at a higher risk of thrombotic vascular

events. Antiplatelet or anticoagulant therapy is not recommended unless indicated for another reason.⁹²

Management of kidney disease is typically not an issue with GCA, although rare patients have ischemic kidney manifestations. Renovascular hypertension is the major kidney problem in Takayasu arteritis.^{88,89} When bilateral renal artery involvement occurs, angiotensin-converting enzyme (ACE) inhibitors may precipitate kidney failure in patients with Takayasu arteritis.⁹⁵ When medical management fails, the renovascular hypertension in patients with Takayasu arteritis may be controlled by vascular surgery or endovascular angioplasty.^{88,89} Reconstructive vascular surgery should be performed during a quiescent phase of the disease.⁹⁰ The management of renovascular hypertension is discussed in [Chapter 43](#).

COVID AND MANAGEMENT OF VASCULITIS

Although only case reports exist on development of vasculitis in patients with COVID-19,^{96–98} the impact is substantial. There are scarce data on treatment in patients with concurrent vasculitis and COVID-19, and controversy on whether immunosuppression is protective or harmful. In some cases immunosuppression was discontinued during active infection,⁹⁹ and steroids were used for treatment in others.⁹⁸ It is unclear if one immunosuppression is superior or if rituximab is safe in patients with vasculitis flare and COVID-19. The question about vaccination is unanswered. How does the COVID-19 vaccine affect management, especially around time of vaccination? Some groups suggest avoiding overimmunosuppression,¹⁰⁰ but the question remains whether we are adequately treating our patients with vasculitis.

SELF-ASSESSMENT QUESTIONS

- In a patient who has systemic vasculitis affecting multiple organs, a kidney biopsy showing GN that is a component of the systemic vasculitis is definitive evidence that the systemic vasculitis is a:
 - large-vessel vasculitis.
 - small-vessel vasculitis.
 - medium-vessel vasculitis.
 - variable-vessel vasculitis.
- If pathologic and serologic studies in a patient with crescentic GN are diagnostic for anti-GBM disease, ANCA testing is:
 - unnecessary because it is unlikely to be positive.
 - unnecessary because it would not change the prognosis if positive.
 - necessary because it is positive in one-fourth to one-third of patients and changes the prognosis.
 - necessary because a positive ANCA result rules out anti-GBM disease.
- Which of the following is *most* common in patients with PAN?
 - Glomerulonephritis
 - Pulmonary hemorrhage
 - Polymyalgia rheumatica
 - Hepatitis B infection
- Which of the following is the *most* common clinical sign or symptom caused by Takayasu arteritis?
 - Flank pain caused by kidney infarction
 - Acute kidney failure
 - Hematuria
 - Hypertension
- Is targeted B-cell therapy with rituximab an acceptable component of induction therapy for ANCA-associated vasculitis and glomerulonephritis?
 - Yes, because induction with rituximab has fewer adverse events than induction with cyclophosphamide.
 - Yes, because clinical trials have shown noninferiority between induction therapy with rituximab compared with cyclophosphamide.
 - No, because clinical trials have shown that induction therapy with rituximab is inferior to induction with cyclophosphamide.
 - No, because rituximab is not approved by the FDA for induction therapy.

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